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Project Title:	Visual Orientation, Navigation, and Spatial Memory Countermeasures		
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Program/Discipline:	NSBRI Teams		
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Space Biology Special Category:	None		
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Harris, Laurence (York University) Liu, Andrew (Massachusetts Institute of Technology) Jenkin, Michael (York University) Aoki, Hirofumi (Massachusetts Institute of Technology)		
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Task Description:	How should spacecraft designers configure interior architectural features, work areas, and the relative orientations of adjacent or docked spacecraft modules to minimize spatial disorientation problems in 0-g ? Current NASA standards offer little guidance. Can virtual reality (VR) training techniques – which astronauts currently use to plan their spacewalks - also be used to reduce the incidence of visual reorientation and inversion illusions while working inside the spacecraft? Is a fully immersive VR system needed for such training, or could simpler, portable training systems be used? Can individual performance on operationally relevant 3D orientation, navigation, and teleoperation tasks be predicted based on simple tests of individual mental rotation and perspective taking skills? What is the best way to assess and control the direction of the perceptual vertical in an environment where there is no gravitational “down”? Can head movement contingent instability of the perceived visual world (“oscillopsia”), experienced by most returning astronauts, be quantified using visual feedback techniques? Answers can be directly applied to the design of the NASA CEV, Lunar Surface Access Module and eventually the Mars Transfer Hab interiors, to the physical arrangement of ground simulators and to the development of VR based techniques for preflight orientation and navigation training for astronauts. Our specific aims were: 1) To quantify how environmental geometric frame and object polarity cues determine human visual orientation, to support engineering and design of spacecraft work areas. 2) To develop reliable means for quantifying head-movement-contingent oscillopsia. 3) To determine whether preflight virtual reality techniques can improve astronaut 3D spatial memory and navigation abilities by reducing direction vertigo, and teaching ISS configuration and emergency egress routes. 4) To improve astronaut teleoperation performance by taking into account the mental object rotation and perspective taking abilities of individuals while training and during operations. By June 2007, we completed all 4 specific aims and all 6 experimental series originally proposed. York studied 1) visual frame and polarity effects in tilted rooms and in an immersive visual virtual environment (IVY), examining the effect of room aspect ratio and observer field of view, 2) the perceptual upright as measured using a new OCHART method ("p" vs. "d" letter recognition) and analyzed results using a linear vector summation model, and 3) quantified oscillopsia during Coriolis stimulation using a new visual feedback technique. The “p/d” method provides us with a way of assessing the perceptual vertical without requiring the subject to make a judgment of tilt with respect to the gravitational vertical – a constraint that has confounded many previous investigations of perceived vertical in ground and 0-g experiments (e.g., Witkin, Mittlestaedt, Howard, Oman). Experiments in IVY, manipulating the floor/ceiling aspect ratio of simple frame interiors, demonstrated that the surface perceived to be "the floor" depends on the aspect ratio in a predictable way that could be mathematically modeled. Several additional experiments were also performed. One showed that the strength of the “levitation” visual reorientation illusion depends on scene content (scene viewed), rather than geometric field of view (view seen). Another showed that the weighting of visual and non-visual cues for orientation was affected by Parkinsonism. MIT completed a series of 4 "relearning reoriented spacecraft modules" experiments, designed to simulate the training experience of astronauts who learn the interiors of individual spacecraft modules in a locally upright configuration in ground simulators, but who have to make spatial judgments when the modules are assembled in a different “flight” configuration. We showed that subjects remember each module in a visually upright, canonical orientation, and therefore had to make mental rotations in order to inter-relate the two modules. This year MIT tested different “flight” configurations, and found that performance was best when visual verticals were co-aligned, intermediate for 180 deg orientations, and worst when modules were rotated through 90 degrees. Our results account for the visualization difficulties and disorientation previously reported by Apollo, Mir and ISS astronauts when transiting certain areas of their spacecraft. The result could be easily translated into a design standard for space stations and docked vehicle operations. MIT also completed two “ISS emergency egress training” studies of 3D, 6 degree of freedom navigation performance, quantifying the effect of training in a locally vs. globally “upright” configuration, with and without smoke obscuration. Most subjects learned quickly, but performance correlated with individual 3D mental rotation and perspective taking skills. This study, led by Dr. Aoki, won the 2007 Young Investigator award from the Aerospace Medical Association's Space Medicine Branch. This year we also compared performance of subjects trained using with a non-immersive laptop display with a similar sized group tested last year using an immersive display. Although immersive displays better simulate the vestibular and haptic cues required to orient spatially, our subjects performed almost as well using the laptop. Finally, as planned, MIT completed development of a space telerobotic training simulator, and showed that individual mental rotation and persepctive taking abilities influence performance during training. Results of the York and MIT studies have been presented at several international meetings and full manuscripts have been published or are currently in submission. Dr. Oman also published a review article on visual orientation in microgravity which summarizes our research in a broader context.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results support the development of sensorimotor countermeasures for spatial orientation, navigation, and spatial memory difficulties among astronauts, and the design of future vehicles, including the CEV, Lunar Surface Access Module and later the Mars Transit Hab. Results have been published in scientific journals (e.g. ASEM, J. Vestib. Res, Habitation), and will be in the NASA Human Integration Design Handbook. Our results also pertain to human health on Earth, for example: origins and assessment of oscillopsia, and disorientation, spatial memory and navigation problems in vestibular, Alzheimer's and Multiple Sclerosis patients. Our results also pertain to the interior design of buildings to reduce disorientation by providing strong visual cues for orientation in both the vertical (to reduce falls in the elderly on stairs) and gravitational horizontal planes (e.g. the origins of “wrong door” phenomena in buildings and “geographic disorientation” in cities, and among sport orienteers, and in the design of visual cueing systems for civil and military flight simulators). To the extent that disorientation is reduced, motion sickness will also be alleviated.

Task Progress:	At York, experiments in a tumbled room and in parabolic flight demonstrated that the direction of the perceptual vertical can be reliably assessed (and mathematically modeled) by having subjects distinguish the identity of a character the identity of which depended on its perceived orientation (e.g., “p” vs. “d”), rather than by manually rotating a visual target or manual rod. The latter methods require the subject to make a judgment of the direction of the perceived gravitational vertical – a task that makes little sense in weightlessness. A new "oblique" method for measurement of the subjective vertical was also developed. Results from a second experiment suggest that the strength of the “levitation” visual reorientation illusion depends on scene content (scene viewed), rather than geometric field of view (view seen). Experiments in the York 6-walled immersive virtual environment (IVY) manipulating the floor/ceiling aspect ratio of simple frame interiors demonstrated that the surface perceived as "the floor" depends on aspect ratio in a predictable, quantifiable manner. This year we also began experiments to see if head-movement-contingent instability of the perceived visual world (“oscillopsia” a phenomenon experienced by most returning astronauts) could be quantified using visual feedback techniques. As a surrogate stimulus for an astronaut’s return to Earth’s gravity we used vestibular Coriolis illusions by having normal subjects make out-of-rotation-plane head movements in a rotating chair. At MIT, our Remote Manipulation Workstation system developed by Dr. Liu was completed, and Alejandra Menchaca-Brandan completed two studies on the influence of perspective-taking and mental rotation abilities on performance during simulated space tele-operation training. Performance was shown to correlate with perspective taking and mental rotation tests. Results formed the basis for a successful NSBRI RFA0701 proposal on teleoperation training. In the area of spatial memory, Claire Cizaire studied of how subjects learn to judge the relative orientation of two docked modules with locally consistent but incongruently aligned interior visual verticals have shown that subjects naturally tend to remember each module in a visually upright orientation. Individual performance was found to correlate with mental rotation and perspective taking skills, and also to depend on relative body orientation, and the relative orientation of the modules. In the area of navigation performance, Dr. Hirofumi Aoki of MIT concluded our studies of 3D, 6 degree of freedom navigation in a simulated ISS emergency egress task. Last year, we studied the effect of training in a locally vs. globally “upright” configuration, with and without smoke obscuration. This year, we prepared results for publication, and also repeated the study with a second group of 36 subjects comparing performance of subjects trained with a non-immersive (laptop) based version of the task.

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